Department of Mathematics
IIT Hyderabad
MA1110: Calculus I

Credits: 1

Fall 2026

Assignment 2

1. Let $f : \mathbb{R} \to \mathbb{R}$ be an additive continuous function, that is, $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Which of the following must be true?

   (A) $f(x) = x^2$ for all $x \in \mathbb{R}$.

   (B) There exists $\lambda \in \mathbb{R}$ such that $f(x) = \lambda x$ for all $x \in \mathbb{R}$.

   (C) $f(x) = |x|$ for all $x \in \mathbb{R}$.

   (D) There exists $\lambda \in \mathbb{R}$ such that $f(x) = \lambda x$ for all $x \in \mathbb{Q}$.

2. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Suppose that $f(x) = 0$ for all $x \in \mathbb{Q}$. Then $f(x) = 0$ for all $x \in \mathbb{R}$.

3. Give two examples of functions $f : \mathbb{R} \to \mathbb{R}$ such that $|f|$ is continuous but f is not continuous at any point in $\mathbb{R}$.

4. Suppose f is a real function defined on $\mathbb{R}$ which satisfies

$$\lim_{h \to 0} [f(x+h) - f(x-h)] = 0 \quad \text{for every } x \in \mathbb{R}.$$

   Does this imply that f is continuous?

5. Let $f : \mathbb{R} \to \mathbb{R}$ be a multiplicative function. (That is, $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$.) Assume that f is continuous at 0. Show that f is continuous on $\mathbb{R}$.

6. Let

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

   Show that f is continuous only at $x = 0$, and that f is differentiable at no point of $\mathbb{R}$.

7. Let $\alpha, \beta \in \mathbb{R}$ and define the function $f_{\alpha,\beta} : [0, \infty) \to \mathbb{R}$ by

$$f_{\alpha,\beta}(x) = \begin{cases} x^{\alpha} \sin\left(\dfrac{1}{x^{\beta}}\right), & x > 0, \\ 0, & x = 0. \end{cases}$$

   (a) For which values of α, β is $f_{\alpha,\beta}$ continuous at $x = 0$?

   (b) For which values of α, β is $f_{\alpha,\beta}$ differentiable at $x = 0$?

   (c) For which values of α, β is $f'_{\alpha,\beta}(0)$ continuous?

8. Suppose f is differentiable in (a, b).

   (a) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.

   (b) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.

   (c) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

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9. (a) If a function is continuous but not differentiable at one point inside (a, b), can the Mean Value Theorem still be applied? Discuss with an example.
   (b) Suppose $f(x) = |x|$ on $[-1, 1]$. Why does the Mean Value Theorem fail here? Which condition is violated?
   (c) Is it possible for the point c from the Mean Value Theorem to be unique? Under what conditions?

10. Let
$$f(x) = x^{2026} - x \sin x - \cos x.$$
Which of the following statements is sufficient to prove that f vanishes at exactly two points in $\mathbb{R}$?
   (A) f is even, $f(0) < 0$, and $f(x) > 0$ for all $x \geq 2$.
   (B) f is even, $f(1) < 0 < f(2)$, and $f'(x) > 0$ for all $x > 0$.
   (C) f is even, $f(1) < 0 < f(2)$, and f' has exactly one zero in $(0, \infty)$.
   (D) f is continuous and $f(1) < 0 < f(2)$.

11. Suppose f is a real function on $(-\infty, \infty)$. Call x a fixed point of f if $f(x) = x$.
   (a) If f is differentiable and $f'(t) \neq 1$ for every real t, prove that f has at most one fixed point.
   (b) Show that the function f defined by
$$f(t) = t + (1 + e^t)^{-1}$$
   has no fixed point, although $0 < f'(t) < 1$ for all real t.
   (c) However, if there is a constant $A < 1$ such that $|f'(t)| \leq A$ for all real t, prove that a fixed point x of f exists, and that $x = \lim x_n$, where x_1 is an arbitrary real number and
$$x_{n+1} = f(x_n)$$
   for $n = 1, 2, 3, \ldots$.

12. If
$$C_0 + \frac{C_1}{2} + \cdots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0,$$
where $C_0, \ldots, C_n$ are real constants, prove that the equation
$$C_0 + C_1 x + \cdots + C_{n-1} x^{n-1} + C_n x^n = 0$$
has at least one real root between 0 and 1.

13. If $f(x) = |x|^3$, compute $f'(x)$, $f''(x)$ for all real x, and show that $f^{(3)}(0)$ does not exist.

14. Let J be an interval and $f, g : J \to \mathbb{R}$ be two differentiable functions such that $f(a) = f(b) = 0$ for two points a and b in J. Show that there exists a point c in (a, b) such that
$$f'(c) + f(c)g'(c) = 0.$$

15. Assume f has a finite derivative in some interval $(a, +\infty)$.
   a) If $f(x) \to 1$ and $f'(x) \to c$ as $x \to +\infty$, prove that $c = 0$.
   b) If $f'(x) \to 1$ as $x \to +\infty$, prove that $f(x)/x \to 1$ as $x \to +\infty$.
   c) If $f'(x) \to 0$ as $x \to +\infty$, prove that $f(x)/x \to 0$ as $x \to +\infty$.

16. Suppose f is defined and differentiable for every $x > 0$, and $f'(x) \to 0$ as $x \to +\infty$. Put $g(x) = f(x+1) - f(x)$. Then prove or disprove that $g(x) \to 0$ as $x \to +\infty$.

17. Suppose the function $f : \mathbb{R} \to \mathbb{R}$ has left and right derivatives at 0. Then, at $x = 0$,

  (a) f must be continuous but may not be differentiable.
  (b) f need not be continuous but must be left continuous or right continuous.
  (c) f must be differentiable.
  (d) if f is continuous then f must be differentiable.

18. What are the following functions $f : \mathbb{R} \to \mathbb{R}$ differentiable at 0 ?

  (A) $f(x) = |x|$.
  (B) $f(x) = |x|^3$.
  (C) $f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0. \end{cases}$
  (D) $f = g^{-1}$ where $g(x) = x^3 + 2x + 3$.

19. Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. Then select the correct options.

  (A) If $A = (a, b)$ and f is continuous and bounded above, then there is $c \in (a, b)$ such that $f(c) = \sup_{x \in (a,b)} f(x)$.
  (B) If $A = \bigsqcup_{n=1}^{\infty} [n - 1/2, n + 1/2]$ and f is continuous on A then f is bounded on A.
  (C) If $A = [a, b]$, then f is continuous implies the intermediate value theorem holds.
  (D) If $A = [1, 2] \cup [3, 4]$ and range of f is A then there is a c in A such that $f(c) = c$.

20. Check whether the limit exists or not. Justify your answer.

  (i) $\lim_{x \to 0} x \cos \frac{1}{x}$.
  (ii) $\lim\limits_{x \to 0} \frac{x^2 \sin(1/x)}{\sin x}$
  (iii) $\lim\limits_{x \to 0} (1 + 2x)^{(x+3)/x}$
  (iv) $\lim\limits_{x \to \infty} x(1 + \sin x)$

21. Prove that the equation
$$(x - 1)^3 + (x - 2)^3 + (x - 3)^3 + (x - 4)^3 = 0$$
can have only one real root.

22. Let $f : [0, 2] \to \mathbb{R}$ be differentiable on $(0, 2)$ and $f(0) = 0,\ f(1) = 4,\ f(2) = 2$. Then prove that there is a $c \in (0, 2)$ such that
$$f'(c) = 0.$$

23. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that, for some constant $C > 0$,
$$|f(x) - f(y)| \leq C|x - y|^2, \qquad \text{for all } x, y \in \mathbb{R}.$$
Prove that f is a constant function.

24. Find f' where $f : \mathbb{R} \to \mathbb{R}$ is given by
$$f(x) = |x| + |x - 1| + |x - 2|.$$

25. Let $f : [0, 3] \to \mathbb{R}$ be defined as
$$f(x) = \begin{cases} x, & \text{for } 0 \leq x \leq 1, \\ 2 - x^2, & \text{for } 1 \leq x \leq 2, \\ x - x^2, & \text{for } 2 \leq x \leq 3. \end{cases}$$
Check for differentiability of the function f at each point $x \in [0, 3]$.

26. Let $f : [a, b] \to \mathbb{R}$ be continuous. Then show that for any $x_1, x_2, x_3 \in [a, b]$, there is $c \in [a, b]$ such that

$$f(c) = \frac{1}{3}\big[f(x_1) + f(x_2) + f(x_3)\big].$$

27. Let $f : [a, b] \to \mathbb{R}$ be continuous such that $f(x) > 0$ for all $x \in [a, b]$. Then prove that there is $\alpha > 0$ in $\mathbb{R}$ such that

$$f(x) > \alpha \qquad \text{for all } x \in [a, b].$$

28. Give two different examples of functions $f : [a, b] \to \mathbb{R}$ which have the intermediate value property in $[a, b]$ but are not continuous on $[a, b]$.

29. Let $f : (-1, 1) \to \mathbb{R}$ be continuous and

$$f(x) = f(x^2).$$

Then prove that f is constant on $(-1, 1)$.

30. If $f'(x)$ exists on $(0, 1)$, then show by Mean Value Theorem that the equation

$$3x^2[f(1) - f(0)] = f'(x)$$

has at least one solution in $(0, 1)$.